

Proposal for Independent Department Project Repository System

Batch-wise GitHub Repository Structure for Student Team Projects

Prepared by: Vaishnav Venugopal

June 10, 2026

Submitted for Review and Recommendation

Contents

1	Introduction	5
2	Problem Statement	5
3	Main Objective	5
4	Nature of the Repository	6
5	Why Batch-wise Structure is Required	6
5.1	Example	7
6	Proposed Repository Structure	7
7	Folder Explanation	8
7.1	README.md	8
7.2	CONTRIBUTING.md	9
7.3	SUBMISSION_GUIDELINES.md	9
7.4	templates/	9
7.5	batches/	9
7.6	archived-projects/	9
8	Student Project Folder Format	9
8.1	Team Project Format	10
8.2	Example	10
9	Required Software Project Folder Structure	10
10	Suggested Electronics or Electrical Project Folder Structure	10
11	Purpose of Each Software Project Folder	11
12	Purpose of Each Electronics or Electrical Project Folder	12
13	README Requirements for Each Project	13
14	Project Report Upload Suggestion	14
15	Git Commands for Accessing and Uploading Projects	14
15.1	Clone the Repository	14
15.2	Enter the Repository Folder	15
15.3	Create a New Branch	15
15.4	Add Project Files	15

15.5 Check File Status	15
15.6 Add Files to Git	15
15.7 Commit Changes	15
15.8 Push Changes	15
16 Student Upload Workflow	15
17 Example Upload	16
18 Access and Recommendation by Faculty	17
19 GitHub Pull Request Template	17
20 GitHub Actions for Basic Checks	18
21 Advantages of the Proposed System	18
22 Suggested Repository Management Model	19
23 Recommended Access Model	19
24 Rules for Students	20
25 Conclusion	20
26 Final Recommendation	21

1 Introduction

This document proposes the creation of an independent department project repository system using GitHub. The purpose of this system is to store, organize, maintain, and showcase student academic projects in a clean and scalable manner.

The repository is designed to support different batches, semesters, subjects, and team projects. It will allow students to independently upload and maintain their projects while also allowing faculty members to access, view, suggest improvements, and recommend the system to students.

2 Problem Statement

Student project files are often shared through different platforms such as Google Drive, email, WhatsApp, pen drives, or local folders. This can create several problems:

- Project files may become scattered across different platforms.
- Students may find it difficult to maintain their work properly.
- Previous batch projects may not be easily available for reference.
- There may be no standard format for project submission or storage.
- Code, reports, screenshots, circuit diagrams, equipment lists, and presentations may not be stored together.
- Useful student projects may be lost after semester completion.

To solve this, a structured GitHub-based independent department repository system is proposed.

3 Main Objective

The main objective is to create a project hub where students can independently upload their academic projects based on:

- Batch
- Semester
- Subject
- Team project

This structure will help students maintain their projects professionally and allow faculty members to access or recommend the system when needed.

4 Nature of the Repository

This repository is proposed as an independent student-driven system. It is not intended to be a compulsory faculty-controlled submission platform.

Faculty members may:

- Access the repository.
- View student projects.
- Recommend students to use the repository.
- Suggest improvements.
- Use the repository as a reference collection.
- Encourage students to maintain projects professionally.

Students may:

- Upload their team projects.
- Maintain project documentation.
- Store source code, reports, screenshots, presentations, and related files.
- Use previous projects for reference.
- Build a professional GitHub profile through contributions.

Repository maintainers may update the structure, add new batches, add new subjects, improve documentation, and maintain the repository when required.

5 Why Batch-wise Structure is Required

A normal year-wise structure such as Year 1, Year 2, and Year 3 is not ideal because students keep moving from one semester to the next.

For example, a student who is currently in Semester 3 will later move to Semester 4, Semester 5, and Semester 6. If the repository is only organized as Year 1 or Year 2, it may become confusing over time.

Therefore, the proposed structure is batch-wise.

5.1 Example

A team from Batch 2027 currently studying Semester 5 should upload DBMS projects under:

```
batches/batch-2027/semester-5/dbms/
```

When the same batch moves to Semester 6, their new projects should be uploaded under:

```
batches/batch-2027/semester-6/final-year-projects/
```

This method keeps the academic progress of each batch clearly organized.

6 Proposed Repository Structure

The proposed GitHub repository can be named:

```
department-project-hub
```

The main structure will be:

```
department-project-hub/  
  README.md  
  CONTRIBUTING.md  
  SUBMISSION_GUIDELINES.md  
  CODE_OF_CONDUCT.md  
  LICENSE  
  .gitignore  
  .github/  
  ISSUE_TEMPLATE/  
    project-upload-request.yml  
    bug-report.yml  
    documentation-request.yml  
  PULL_REQUEST_TEMPLATE.md  
  workflows/  
    basic-checks.yml  
  templates/  
    software-project-template/  
    electronics-project-template/  
    mini-project-template/  
    final-year-project-template/  
  batches/  
    batch-2027/  
      semester-1/  
      semester-2/
```

```
semester-3/  
semester-4/  
semester-5/  
  dbms/  
  web-development/  
  software-engineering/  
  electronics-projects/  
semester-6/  
batch-2028/  
batch-2029/  
archived-projects/  
  batch-2024/  
  batch-2025/  
  batch-2026/
```

7 Folder Explanation

7.1 README.md

The main README file explains the purpose of the repository, the folder structure, student upload process, usage instructions, and project documentation rules.

Each project README should clearly mention:

- Project title.
- Team name and team members.
- Batch, semester, and subject.
- Project description.
- Tools and technologies used for creation.
- Programming languages, frameworks, libraries, or platforms used.
- Sources or references used for coding and learning.
- Setup and installation instructions.
- Output screenshots or demo explanation.
- Project report PDF, if available.
- Google Drive report link, if the report file is large.
- Future improvement ideas, if any.

7.2 CONTRIBUTING.md

This file explains how students can contribute to the repository. It includes instructions for cloning the repository, creating branches, adding project files, committing changes, pushing changes, and opening pull requests.

7.3 SUBMISSION_GUIDELINES.md

This file explains the required project upload format. It includes rules for folder naming, required files, screenshots, reports, presentations, installation instructions, tools used, and references used.

7.4 templates/

The templates folder contains standard project templates that students can copy before uploading their own project.

There can be separate templates for:

- Software projects.
- Electronics or electrical projects.
- Mini projects.
- Final-year projects.

7.5 batches/

This is the most important folder. It contains separate folders for each batch.

Example:

```
batches/batch-2027/  
batches/batch-2028/  
batches/batch-2029/
```

7.6 archived-projects/

This folder stores completed or older batch projects for future reference.

8 Student Project Folder Format

Each team should create a project folder using a standard naming format.

8.1 Team Project Format

```
team-01-project-name  
team-02-project-name  
team-03-project-name
```

8.2 Example

```
batches/batch-2027/semester-5/dbms/team-01-library-management/
```

Individual project folders are not included in this structure because the repository is mainly intended for team-based academic project uploads.

9 Required Software Project Folder Structure

Every software project should preferably follow this format:

```
project-name/  
  README.md  
  src/  
  docs/  
  screenshots/  
  reports/  
  presentation/  
  installation-guide.md  
  output-demo.md
```

10 Suggested Electronics or Electrical Project Folder Structure

For electronics, electrical, embedded systems, IoT, robotics, or hardware-based projects, the following structure is suggested:

```
project-name/  
  README.md  
  code/  
  circuit-diagrams/  
  component-list/  
  equipment-used/  
  datasheets/  
  simulations/
```

```
screenshots/  
photos/  
reports/  
presentation/  
working-video-link.md  
setup-and-testing.md  
output-demo.md
```

If available, students are encouraged to upload:

- Project report as a PDF file.
- Google Drive link for the project report if the PDF file is large.
- Circuit diagram.
- Block diagram.
- Components used.
- Equipment used.
- Simulation files.
- Code for microcontrollers or development boards.
- Datasheets of important components.
- Photos of hardware setup.
- Testing procedure.
- Output result.
- Working video link.

11 Purpose of Each Software Project Folder

Folder/File	Purpose
README.md	Contains project title, description, team members, tools used, references used, features, technologies used, setup details, and report link if required.
src/	Contains the source code of the project.
docs/	Contains additional documentation.

screenshots/	Contains screenshots of the project output or user interface.
reports/	Contains project report PDF, if available. If the report file is large, a Google Drive link can be added in the README or in <code>report-link.md</code> .
presentation/	Contains presentation files such as PPT or PDF, if available.
installation-guide.md	Explains how to install and run the project.
output-demo.md	Explains the output, demo steps, or final result of the project.

12 Purpose of Each Electronics or Electrical Project Folder

Folder/File	Purpose
README.md	Contains project title, description, team members, tools used, components used, equipment used, references, and report link if required.
code/	Contains Arduino, ESP, Raspberry Pi, microcontroller, simulation, or other project code.
circuit-diagrams/	Contains circuit diagrams, block diagrams, wiring diagrams, or PCB design files.
component-list/	Contains the list of electronic components used in the project.
equipment-used/	Contains details of equipment used, such as multimeter, oscilloscope, power supply, sensors, modules, boards, and testing tools.
datasheets/	Contains datasheets of important components, if available.
simulations/	Contains simulation files from tools such as Proteus, Tinkercad, Multisim, MATLAB, or similar tools, if available.
screenshots/	Contains screenshots of software output, simulation output, or serial monitor output.
photos/	Contains photos of the hardware setup or final working model.

reports/	Contains the project report PDF, if available. If the report file is large, a Google Drive link can be added in the README or in <code>report-link.md</code> .
presentation/	Contains presentation files such as PPT or PDF, if available.
working-video-link.md	Contains the link to a working video demo, if available.
setup-and-testing.md	Explains hardware setup, testing process, observations, and troubleshooting steps.
output-demo.md	Explains the final output and working result of the project.

13 README Requirements for Each Project

Each project README should include the following details:

- Project title.
- Batch, semester, and subject.
- Team name and team members.
- Short project description.
- Problem statement.
- Features or working principle.
- Tools used for creation.
- Technologies, programming languages, frameworks, or hardware platforms used.
- Sources or references used for coding, circuit design, learning, or implementation.
- Installation or setup instructions.
- Steps to run or test the project.
- Screenshots, photos, or demo output.
- Project report PDF, if available.
- Google Drive link for the project report if the PDF file is too large to upload directly to GitHub.
- Presentation details, if available.

- Future improvements, if any.

For electronics or electrical projects, the README should also include:

- Components used.
- Equipment used.
- Circuit diagram or block diagram details.
- Power supply details.
- Software or simulation tools used.
- Testing method.
- Final output or observation.
- Project report PDF or Google Drive report link, if available.

14 Project Report Upload Suggestion

When adding a project to GitHub, students are encouraged to upload their project report as a PDF file if available. The report can be placed inside the `reports/` folder.

Example:

```
reports/project-report.pdf
```

If the project report is too large to upload directly to GitHub, students can upload the report to Google Drive and add the shareable link in the project README file or in a separate file named:

```
report-link.md
```

The Google Drive link should have proper sharing permission so that others can view the report.

15 Git Commands for Accessing and Uploading Projects

Students can access and upload projects using Git commands.

15.1 Clone the Repository

```
git clone https://github.com/organization-name/department-project-hub.git
```

15.2 Enter the Repository Folder

```
cd department-project-hub
```

15.3 Create a New Branch

```
git checkout -b batch-2027-sem5-dbms-team-01
```

15.4 Add Project Files

Students should place their project inside the correct path.

Example:

```
batches/batch-2027/semester-5/dbms/team-01-library-management/
```

15.5 Check File Status

```
git status
```

15.6 Add Files to Git

```
git add .
```

15.7 Commit Changes

```
git commit -m "Add DBMS project for Batch 2027 Semester 5"
```

15.8 Push Changes

```
git push origin batch-2027-sem5-dbms-team-01
```

After pushing, students can open a pull request on GitHub to request merging their project into the main repository.

16 Student Upload Workflow

The proposed student upload workflow is:

1. Students open or clone the department project repository.

2. Students create a new branch for their team project.
3. Students select the correct batch folder.
4. Students select the correct semester folder.
5. Students select the correct subject folder.
6. Students create a team project folder using the correct naming format.
7. Students add source code, documentation, screenshots, reports, presentation, and other required files.
8. Students mention tools used and references used in the project README.
9. Students upload the project report as a PDF file if available.
10. If the project report file is large, students can upload the report to Google Drive and add the shareable Google Drive link inside the project README or a separate file named `report-link.md`.
11. Students commit and push the changes.
12. Students create a pull request or request the repository maintainer to merge the project.
13. The project becomes available for future reference.

17 Example Upload

For a Batch 2027 team in Semester 5 uploading a DBMS project titled Library Management System, the correct path will be:

```
batches/batch-2027/semester-5/dbms/team-01-library-management/
```

The folder should contain:

```
team-01-library-management/  
README.md  
src/  
docs/  
screenshots/  
reports/  
presentation/  
installation-guide.md  
output-demo.md
```

If the report is available, it can be uploaded as:

```
team-01-library-management/reports/project-report.pdf
```

If the report is large, the team can add a Google Drive link inside:

```
team-01-library-management/report-link.md
```

18 Access and Recommendation by Faculty

Although the repository is independent and student-driven, faculty members can still access and recommend it.

Faculty involvement may include:

- Recommending students to upload projects in the correct structure.
- Accessing projects for reference.
- Suggesting improvements to documentation or organization.
- Encouraging better project maintenance.
- Using selected projects as examples for future students.
- Helping students understand professional version control practices.

The system does not require compulsory faculty approval for every upload unless the department or repository maintainers decide to add such a rule later.

19 GitHub Pull Request Template

A pull request template can be included so that every student upload contains the required details.

The pull request form can ask for:

- Team name.
- Team members.
- Batch year.
- Current semester.
- Subject.
- Project title.

- Project path.
- Project type.
- Tools used.
- References or sources used.
- Project report PDF or Google Drive report link, if available.
- Upload checklist.

20 GitHub Actions for Basic Checks

A basic GitHub Actions workflow can be added to check common upload issues.

The workflow can check:

- Whether README.md exists in uploaded project folders.
- Whether empty project folders are uploaded.
- Whether files are placed under the correct path.
- Whether unnecessary files such as node_modules, build, dist, .env, .DS_Store, and __pycache__ are committed.
- Whether helpful messages are shown when a problem is found.

21 Advantages of the Proposed System

- Centralized storage of student team projects.
- Easy tracking of projects by batch, semester, and subject.
- Independent project maintenance by students.
- Better visibility of student work.
- Standard project upload format.
- Previous batch projects can be preserved.
- Students can build a professional GitHub contribution history.
- Faculty can access and recommend projects when needed.
- Useful for references, portfolios, and future improvements.

- Reduces confusion caused by scattered files.
- Supports future batches and semesters.
- Supports both software and electronics/electrical projects.
- Allows reports to be stored as PDF files or linked through Google Drive when large.

22 Suggested Repository Management Model

The recommended model is:

GitHub Organization or Shared Repository

Department Project Hub Repository

Batch-wise Folders

Semester-wise Folders

Subject-wise Folders

Team Project Folders

Independent Student Uploads

Faculty Access and Recommendation

23 Recommended Access Model

The department or student maintainers can manage access as follows:

Role	Suggested Access
Repository Maintainer	Full access to maintain structure and merge uploads.
Student Teams	Can upload team projects using branches or pull requests.
Faculty Members	Can access, view, suggest, and recommend the repository.
External Viewers	Read-only access if the repository is public.

24 Rules for Students

Students should follow these rules:

- Upload projects only under the correct batch folder.
- Select the correct semester folder.
- Select the correct subject folder.
- Upload projects using the team project folder format.
- Do not modify or delete other teams' projects.
- Do not upload private keys, passwords, tokens, or API secrets.
- Do not upload unnecessary large files.
- Include proper documentation.
- Mention tools used for creating the project.
- Mention sources or references used for coding, design, circuit creation, or learning.
- Upload the project report as a PDF file if available.
- If the project report is large, upload it to Google Drive and add the shareable link in the README or in a file named `report-link.md`.
- Upload presentation, screenshots, photos, or working video link if available.
- Prefer uploading through Git commands and pull requests.
- Follow academic honesty rules.

25 Conclusion

The proposed batch-wise GitHub department project repository system provides a clean, scalable, and professional way to manage student team projects independently.

Instead of organizing projects only by academic year, the batch-wise method allows each batch to continue from one semester to the next without confusion. It also helps students maintain their projects properly and allows faculty members to access or recommend the repository when needed.

This system can be implemented using GitHub and Git commands. Repository maintainers can update the structure, add batches, add subjects, and improve documentation whenever required.

26 Final Recommendation

It is recommended to create:

- One GitHub Organization or shared GitHub repository for the department project hub.
- One central repository named `department-project-hub`.
- Batch-wise folders inside the repository.
- Semester-wise folders inside each batch.
- Subject-wise folders inside each semester.
- Team project folders inside each subject.
- Software and electronics/electrical project templates.
- Option to upload project reports as PDF files or provide Google Drive links for large reports.
- Faculty access for viewing, suggesting, and recommending the system.

This structure is suitable for current students, future batches, faculty reference, and long-term department project archiving.